

AEC / SDC · SEMESTER III · PROJECT-BASED LEARNING · ONE CREDIT

Complete Community-Project Notes with the Tool It Documents

COMMUNITY PROJECT

Topic 1 — Computer-Based Learning Tool for Mentally Challenged Students
Mangaluru, Dakshina Kannada · Karnataka, India

CONTINUATION COMPANION

Secondary Data · Open-Source Model · District Mapping
English-Kannada Browser Tool · Git and Render Handover

COURSE CODE

1BCP308

Credits	Teaching Hours (L : T : P)	Total Marks	CIE	SEE	Exam Hours	Examination Type
1	0:0:0	100	50	50	03 Hours	Practical

Project application: SAHAYA

Prepared for project-team adaptation and special-educator review
Team / USNs / Guide / School: to be supplied

Format reference: Community-Project-1BCP308-Book.pdf

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No authorship, affiliation, school acceptance or endorsement implied.

20 September 2026 · Engineering prototype with pending field validation

FRONT MATTER

Course at a Glance

Item	Record
Course	Community Project / Societal Project (Project-Based Learning)
Course code	1BCP308 · AEC/SDC · Semester III · 1 credit
Pedagogy ledger	0:0:0:30:30; planned term work 30 h + self-learning 30 h
Project location	Dakshina Kannada district, Karnataka, India; Mangaluru headquarters
Application	Sahaya: English-Kannada foundational practice
POs / SDG	Specific POs 5, 6 · SDG 4
Runtime	Static HTML/CSS/JavaScript; local assets; no learner backend
Secondary sources	UCI handwritten digits; geoBoundaries district geometry; Twemoji graphics
Open-source model	scikit-learn 3-NN reference; browser equivalent for examples
Technical status	Built, tested and captured; local Git initialized; Render Blueprint supplied
Field status	No school survey, interviews or learner trials; Kannada expert review pending
Publication status	No remote Git push or Render account deployment performed

Environment of record

python: 3.13.14; platform: Linux-6.1.158+-x86_64-with-glibc2.41; numpy: 2.3.5; scikit_learn: 1.6.1; node: v20.20.2;
 record_date: 2026-09-20

This report preserves the book's week-led sequence, numbered six-part chapter pattern, code listings, captured-run boxes, checkpoint questions and appendices. It is a newly generated companion, not an alteration of the supplied PDF. Original pilot numbers are not reused.

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Preface

The useful continuation of a community-project book is a tool a team can run, a report it can defend, and a clear boundary around what has not been done. Sahaya provides all three. Its engineering evidence comes from programs executed in this workspace and from attributed open repositories, not from invented stakeholder responses.

Each of the fifteen chapters follows the corresponding Teaching-Learning Process week. Sections retain the reference pattern: Why This Week; The Program (Code); Captured Run (Two-Pass Verified); What the Capture Proves; Checkpoint Questions; and Field Extensions. The listings are the actual executable checks included in the repository.

The code and data are ready for local reproduction. A Git history, static Render Blueprint and CI workflow are supplied. Publication through the user's Git and Render accounts remains an explicit handover step. The live sandbox preview is not described as a public Render deployment.

The title from the syllabus is reproduced exactly in the prescribed topic row. New narrative uses learners with intellectual disabilities or learners requiring individualized support, and treats a special educator—not a model—as the authority on suitability. The generic baseline must be reviewed with a consenting local school before classroom use.

Evidence discipline

MEASURED: source row counts, content totals, model evaluation, automated tests, artifact fingerprints and payload size.

SOURCED: district context, boundary vintage, repository provenance and licences.

PROPOSED: fifteen-week plan, stakeholder protocol, local requirement validation and deployment steps.

NOT CLAIMED: school partnership, learner benefit, diagnosis, Kannada quality certification, all-device accessibility, remote publication or achieved marks.

RESULT: 15 week programs passed; two captures each; byte-identical stdout

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Prescribed Course Content — Topic Row

The column order and requested wording are preserved below. The measurable outcome remains an objective; the phrase “as per the requirements of the special school” is not treated as achieved before a school supplies and approves those requirements.

Sl. No.	Broad Areas/Project titles	Measurable engineering outcome	Specific POs	SDGs
1	Computer-Based Learning Tool for Mentally Challenged Students	Design and develop a web/browserbased learning tool as per the requirements of the special school.	5, 6	4

Assessment arithmetic from the supplied book

Criterion (short label)	CIE	SEE
Problem identification and relevance	8	16
Problem statement, engineering and feasibility	10	20
Stakeholder interaction and data	8	16
Documentation and problem-solving	10	20
Communication and presentation	6	12
Societal contribution, impact and sustainability	8	16
Total	50	100 (scaled to 50)

CIE $\geq 20/50$ for SEE eligibility; SEE $\geq 35/100$ in the examination itself; CIE plus SEE scaled to 50 must total ≥ 40 . These rules and shortened rubric labels are restated from the supplied book, not independently certified as the current university regulation. Full performance-band wording remains in the reference book.

Course-outcome and PO alignment

CO identifiers in the chapter metadata follow the corresponding rows of the reference book’s compliance map. They are traceability labels, not a finding that every course outcome has been attained. PO5: appropriate modern-tool use with limits. PO6: responsible community application, accessibility, privacy and stakeholder involvement. SDG 4: intended alignment with inclusive quality education, not a measured indicator change.

FRONT MATTER

Teaching-Learning Process — Fifteen Weeks

W k	Activity	Output / evidence	Lead role
1	Course introduction, objectives, ethics, conduct, and assessment briefing	Team list, course plan	Coordinator / ethics
2	Community problem identification and topic selection	Shortlisted project topic	Content / accessibility
3	Stakeholder interaction and preliminary survey	Survey notes, interview record (pending)	Engineering / data
4	Problem statement and objective formulation	Problem statement and requirement ledger	QA / documentation
5	Literature/case study review and baseline analysis	Baseline analysis summary	Coordinator / ethics
6	Project planning and role assignment	Work plan / Gantt chart	Content / accessibility
7	Feasibility analysis and solution brainstorming	Solution alternatives sheet	Engineering / data
8	Selection of final solution and design approach	Concept note / design outline	QA / documentation
9	Data processing / calculations / sizing / mapping	Technical analysis sheet	Coordinator / ethics
10	Prototype / model / audit / DPR / app development	Working app / source repository	Content / accessibility
11	Testing, refinement, and implementation planning	Test results / revised output	Engineering / data
12	Field implementation / demonstration / awareness activity	Implementation record (pending)	QA / documentation
13	Documentation of outcomes, feedback, and impact	Draft report / impact limits	Coordinator / ethics
14	Presentation preparation and viva practice	Presentation outline / reproducibility receipts	Content / accessibility
15	Final presentation, submission, and reflection	Final report, deployment handover, viva	Engineering / data

All rows are a proposed schedule, not an attendance record. Two term-work hours and two self-learning hours per row: 30 + 30 = 60 hours. Faculty roles are to guide framing, demonstrate ethical engagement, review feasibility/technical work, facilitate permitted fieldwork and assess evidence. Student owners and reviewers must be assigned locally.

FRONT MATTER

Compliance Maps & Assessment Evidence

Evidence family	Repository location	Meaning / limitation
Requirements and ethics	docs/REQUIREMENTS.md; FIELD_PROTOCOL.md	Generic baseline and blank protocol, not signed school approval
Content and secondary data	public/data; data/raw	Thirty cards; source-derived digits and boundary
Model and implementation	scripts/prepare_data.py; public/engine.js	3-NN reference evaluation and local demonstration
Test receipts	evidence/unit-transcript.txt; browser-results.json	Recorded automation; no learner field trial
Two-pass capture	scripts/collect.py; evidence/cp00..cp14	Fifteen deterministic checks; stdout byte equality
Version control and hosting	Local Git, render.yaml, docs/DEPLOYMENT.md	Prepared handover, not remote deployment
Course reflection and viva	Chapters 13-15; Appendices B-D	Separate evidence from unresolved fieldwork

Measurable engineering outputs

Measure	Captured value
Bilingual cards	30; 20 authored everyday + 10 UCI digit cards
Model source split	3823 train / 1797 test
Reference digit accuracy	$1758/1797 = 97.83\%$
Automated checks	9 Node tests; 19 Chromium checks
Accessibility scan	0 detected axe violations across 4 main views
Local field engagement	0 school surveys / 0 learner trials
Learner improvement	Not measured

Technical metrics are narrow engineering receipts. They cannot be translated into an educational effectiveness percentage or full marks.

FRONT MATTER

List of Contents

Body numbering restarts at 1. Each chapter contains sections .1 through .6 in the reference book's order. Front matter uses roman-numbered footers.

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PART A

METHOD: THE CASE FOR ONE TOOL

Chapters 1–8 · Community context, secondary evidence, requirements, source review, schedule, feasibility and design.

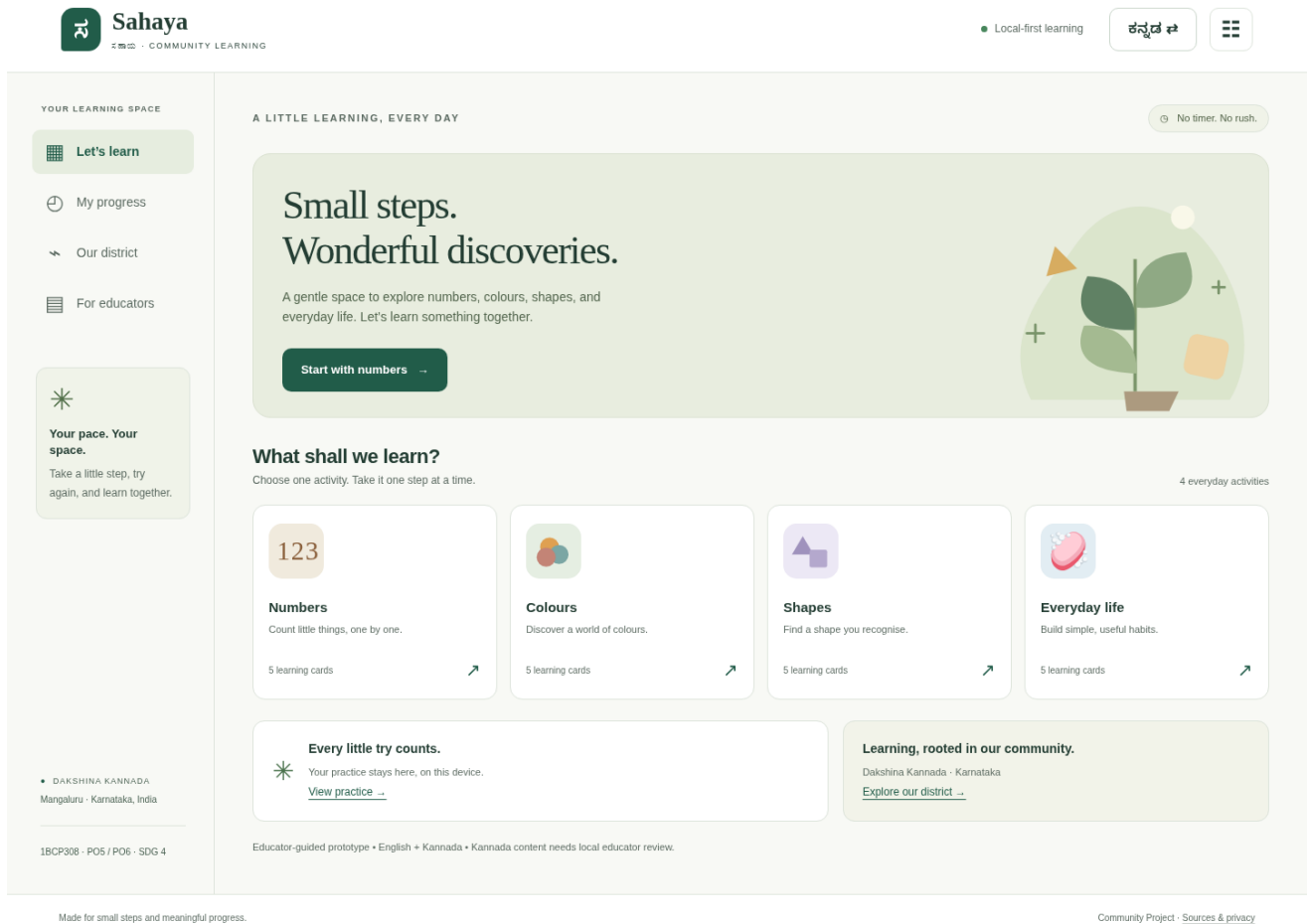


Figure A. Sahaya home screen captured from the running prototype. This is technical evidence, not a photograph of classroom use.

Chapter 1 — The Briefing Ledger: What the Course Actually Pays For

WEEK (TLP ROW) 1 — Course introduction, objectives, ethics, conduct, and assessment briefing

EVIDENCE Team list, course plan

FILES scripts/cp00_week.py; evidence/cp00_week.txt

COURSE OUTCOMES CO1, CO2 (reference alignment only)

1.1 Why This Week

A community project is assessed on more than whether a screen opens. The supplied 1BCP308 book begins with an assessment ledger, and this companion retains that order. The exact requested topic row remains unchanged. New engineering work is described separately from pending school engagement; no marks, approval, or attendance are invented.

1.2 The Program (Code)

The listing below is the file executed for the captured run. Shared I/O helpers are reproduced in Appendix D.

```
from evidence_common import result
cie = [8, 10, 8, 10, 6, 8]
see = [16, 20, 16, 20, 12, 16]
assert sum(cie) == 50 and sum(see) == 100
passing = [raw for raw in range(101)
            if raw >= 35 and 20 + raw / 2 >= 40]
print(f'CIE={sum(cie)}; SEE={sum(see)}; scaled SEE=/50')
print('Gates: CIE >=20/50; SEE >=35/100; total >=40/100')
print(f'At CIE=20, minimum passing raw SEE={min(passing)}')
result(0, 'assessment arithmetic verified; no grade awarded')
```

1.3 Captured Run (Two-Pass Verified)

```
$ python scripts/cp00_week.py
CIE=50; SEE=100; scaled SEE=/50
Gates: CIE >=20/50; SEE >=35/100; total >=40/100
At CIE=20, minimum passing raw SEE=40
CP-00 RESULT: assessment arithmetic verified; no grade awarded
```

1.4 What the Capture Proves

The program reproduces the CIE/SEE weight arithmetic from the supplied book and computes the boundary case at CIE 20. It does not authenticate the current university syllabus or award a grade. Confirm operative assessment rules with the department. PO5 is evidenced by the engineering methods; PO6 requires responsible application and, ultimately, genuine stakeholder engagement.

RESULT (THE PROGRAM'S OWN LINE)

CP-00 RESULT: assessment arithmetic verified; no grade awarded

1.5 Checkpoint Questions

Q1. Why is a working app not sufficient evidence for the whole course?

Model answer. The rubric also covers problem definition, stakeholder interaction, documentation, communication and social benefit. Secondary datasets cannot stand in for school consultation.

Q2. Why keep the historical topic title?

Model answer. The user requested the exact prescribed table format. The row is retained as a quotation, while the new application uses respectful, learner-centred terminology.

1.6 Field Extensions

Enter team members, USNs, guide and institution only after they are supplied. Agree on a school-contact plan and a safe evidence-handling policy. Never add a signature, affiliation or permission number on someone else's behalf.

Chapter 2 — Screening the Topic Against the Ground It Must Land On

WEEK (TLP ROW) 2 — Community problem identification and topic selection

EVIDENCE Shortlisted project topic

FILES scripts/cp01_week.py; evidence/cp01_week.txt

COURSE OUTCOMES CO1, CO5 (reference alignment only)

2.1 Why This Week

The location is Dakshina Kannada district, Karnataka, India, with its headquarters at coastal Mangaluru. The official district administration page supplies this context [S3]. The map is not a decoration drawn from memory: it is an extracted feature from the geoBoundaries India ADM2 release [S2], with the source year and licence kept beside the data.

2.2 The Program (Code)

The listing below is the file executed for the captured run. Shared I/O helpers are reproduced in Appendix D.

```
from evidence_common import load, result
geo = load('public/data/dakshina-kannada.geojson')
meta = load('data/raw/geoboundaries-metadata.json')
assert len(geo['features']) == 1
name = geo['features'][0]['properties']['shapeName']
assert name == 'Dakshina Kannada'
print(f'District={name}; state=Karnataka; country=India')
print('Headquarters=Mangaluru; point approximate, not surveyed')
print('Boundary vintage=' + meta['boundaryYearRepresented'])
print('Boundary licence=' + meta['boundaryLicense'])
result(1, 'one named district source feature verified')
```

2.3 Captured Run (Two-Pass Verified)

```
$ python scripts/cp01_week.py
District=Dakshina Kannada; state=Karnataka; country=India
Headquarters=Mangaluru; point approximate, not surveyed
Boundary vintage=2021
Boundary licence=Open Data Commons Open Database License 1.0
CP-01 RESULT: one named district source feature verified
```

2.4 What the Capture Proves

The check confirms exactly one named district feature and its recorded 2021 vintage. The administrative facts are sourced externally; an assertion alone is not external verification. An approximate Mangaluru city point is a hand-entered orientation annotation, not a surveyed school location. The sea shading and Western Ghats label are schematic context, not additional datasets.

RESULT (THE PROGRAM'S OWN LINE)

CP-01 RESULT: one named district source feature verified

2.5 Checkpoint Questions

Q1. Why does the map say 2021 rather than 2026?

Model answer. 2021 is the represented year in this boundary source. Retrieval in 2026 does not make the geometry a current official boundary.

Q2. Why are special schools not pinned on this map?

Model answer. No authoritative, licensed, validated school-location dataset or partnership list was obtained. Invented locations would mislead educators and examiners.

2.6 Field Extensions

Verify any future school locations with the institution and an authoritative open source. Document consent and accuracy; never display home addresses or individual learner locations.

Chapter 3 — The Baseline Survey: Secondary Evidence, Primary Work Pending

WEEK (TLP ROW) 3 — Stakeholder interaction and preliminary survey

EVIDENCE Survey notes, interview record (pending)

FILES scripts/cp02_week.py; evidence/cp02_week.txt

COURSE OUTCOMES CO1, CO2, CO4 (reference alignment only)

3.1 Why This Week

Week 3 in the reference book concerns stakeholder interaction and a preliminary survey. This delivery has no primary survey. It supplies genuine secondary engineering data instead: the UCI Optical Recognition of Handwritten Digits dataset [S1]. Its rows describe handwritten digit intensities, not local children or educational needs. The field protocol is therefore explicitly marked PENDING.

3.2 The Program (Code)

The listing below is the file executed for the captured run. Shared I/O helpers are reproduced in Appendix D.

```
from evidence_common import ROOT, read, result
counts = {}
for part in ['tra', 'tes']:
    rows = read(f'data/raw/optdigits.{part}').splitlines()
    counts[part] = len(rows)
    assert all(len(row.split(',')) == 65 for row in rows)
assert counts == {'tra': 3823, 'tes': 1797}
assert 'PENDING' in read('docs/FIELD_PROTOCOL.md')
print(f"Secondary digit rows={sum(counts.values())}")
print('Local schools surveyed=0; local learners tested=0')
print('Primary engagement status=PENDING')
result(2, 'secondary data separated from unperformed fieldwork')
```

3.3 Captured Run (Two-Pass Verified)

```
$ python scripts/cp02_week.py
Secondary digit rows=5620
Local schools surveyed=0; local learners tested=0
Primary engagement status=PENDING
CP-02 RESULT: secondary data separated from unperformed fieldwork
```

3.4 What the Capture Proves

The script counts the bundled original training/test files and verifies 65 fields per row: 64 pixel intensities plus the class. It does not reinterpret these rows as survey respondents. UCI documents 43 source writers in distinct training/test groups; they are not participants in this community project. Source data are licensed and attributed, with file fingerprints recorded.

RESULT (THE PROGRAM'S OWN LINE)

CP-02 RESULT: secondary data separated from unperformed fieldwork

3.5 Checkpoint Questions

Q1. Can the dataset prove the special school needs this tool?

Model answer. No. It enables technical prototyping and reproducible model evaluation. School requirements require direct authorized consultation.

Q2. Why not use the reference book's respondent and pilot numbers?

Model answer. Those numbers belong to another supplied artifact and its methodology. They are not observations made for this implementation and are not copied into the evidence ledger.

3.6 Field Extensions

Use the blank observation record in docs/FIELD_PROTOCOL.md. Arrange an educator interview, record actual constraints, and leave all respondent/learner counts at zero until authorized engagement occurs.

Chapter 4 — Freezing the Problem Statement on Evidence Anchors

WEEK (TLP ROW) 4 — Problem statement and objective formulation

EVIDENCE Problem statement and requirement ledger

FILES scripts/cp03_week.py; evidence/cp03_week.txt

COURSE OUTCOMES CO1, CO2 (reference alignment only)

4.1 Why This Week

The baseline problem statement is: develop a browser tool for educator-guided foundational practice in English and Kannada, with understandable feedback, local privacy controls, auditable resources and district context. This is a proposed baseline for local validation—not a claim to have implemented an unnamed school’s actual requirements.

4.2 The Program (Code)

The listing below is the file executed for the captured run. Shared I/O helpers are reproduced in Appendix D.

```
from evidence_common import load, read, result
cards = load('public/data/lessons.json')
text = read('docs/REQUIREMENTS.md')
assert all(f'R{i}' in text for i in range(1, 11))
assert all(c['prompt']['en'] and c['prompt']['kn'] for c in cards)
assert '5, 6' in text and '| 4 |' in text
print('Requirements=R1..R10; bilingual cards=' + str(len(cards)))
print('Specific POs=5,6; SDG=4; school approval=PENDING')
result(3, 'baseline traceability exists; school sign-off not claimed')
```

4.3 Captured Run (Two-Pass Verified)

```
$ python scripts/cp03_week.py
Requirements=R1..R10; bilingual cards=30
Specific P0s=5,6; SDG=4; school approval=PENDING
CP-03 RESULT: baseline traceability exists; school sign-off not claimed
```

4.4 What the Capture Proves

R1-R10 in the requirement ledger link features to evidence and outstanding review. The content has both language fields; that structural check cannot certify translation quality. The learner-facing interface and prompts are bilingual, while technical educator notes and map labels are English-only. School-specific requirements and Kannada specialist review remain release gates.

RESULT (THE PROGRAM'S OWN LINE)

CP-03 RESULT: baseline traceability exists; school sign-off not claimed

4.5 Checkpoint Questions

Q1. What is measurable now?

Model answer. Card count, language-field coverage, data split, model accuracy, tests, artifact hashes and deployment configuration. User benefit is not yet measured.

Q2. What is deliberately outside scope?

Model answer. Diagnosis, automated ability ranking, private learner profiles, cloud tracking, authenticated administration and unconstrained generative content.

4.6 Field Extensions

Review each requirement with a special educator. Replace generic assumptions with signed-off acceptance criteria, then add actual device and access constraints without collecting unnecessary personal information.

Chapter 5 — The Baseline: What Already Exists, and What It Misses

WEEK (TLP ROW) 5 — Literature/case study review and baseline analysis

EVIDENCE Baseline analysis summary

FILES scripts/cp04_week.py; evidence/cp04_week.txt

COURSE OUTCOMES CO1, CO5 (reference alignment only)

5.1 Why This Week

The baseline comparison is intentionally practical. A static application minimizes operational dependencies and keeps practice local. A general-purpose LMS adds accounts and administration; a runtime large language model adds cost, variable content and governance burdens. Printed educator cards remain a useful fallback. These are design trade-offs, not a systematic literature review or a superiority claim.

5.2 The Program (Code)

The listing below is the file executed for the captured run. Shared I/O helpers are reproduced in Appendix D.

```
from evidence_common import read, result
sources = read('docs/SOURCES.md')
for term in ['UCI', 'geoBoundaries', 'Twemoji', 'scikit-learn',
            'IndicTrans2', 'Google Dataset Search']:
    assert term in sources
assert 'Not downloaded or executed' in sources
print('Used: UCI digits, geoBoundaries, Twemoji, scikit-learn')
print('Optional only: IndicTrans2; finder links are discovery aids')
result(4, 'source register and model-use boundaries documented')
```

5.3 Captured Run (Two-Pass Verified)

```
$ python scripts/cp04_week.py
Used: UCI digits, geoBoundaries, Twemoji, scikit-learn
Optional only: IndicTrans2; finder links are discovery aids
CP-04 RESULT: source register and model-use boundaries documented
```

5.4 What the Capture Proves

The source register distinguishes resources actually used from resources merely discoverable. UCI data, geoBoundaries geometry, Twemoji graphics and scikit-learn are integrated. AI4Bharat IndicTrans2 En-Indic distilled 200M [S6] is an optional translation candidate with Kannada support and an MIT licence listed on its model card. It was not downloaded or executed. Finder links do not make every result open-licensed or pedagogically suitable.

RESULT (THE PROGRAM'S OWN LINE)

CP-04 RESULT: source register and model-use boundaries documented

5.5 Checkpoint Questions

Q1. Which open-source model actually runs?

Model answer. The Python scikit-learn 3-nearest-neighbours baseline and a small original browser equivalent. No foundation-model API is called.

Q2. Why list a translation model if it is not integrated?

Model answer. To document a plausible future resource and its limitations transparently. The app must not imply it generated or validated the current Kannada content.

5.6 Field Extensions

Search UCI, Google Dataset Search, Hugging Face, OGD India and Kaggle with a precise question. Record creator, version, licence, population, intended use and representation before selecting a resource.

Chapter 6 — The Week Map: Fifteen Slots, Roles and Review Gates

WEEK (TLP ROW) 6 — Project planning and role assignment

EVIDENCE Work plan / Gantt chart

FILES scripts/cp05_week.py; evidence/cp05_week.txt

COURSE OUTCOMES CO2 (reference alignment only)

6.1 Why This Week

The fifteen-week schedule follows the Teaching-Learning Process order in the supplied book. It is a proposed workload structure, not evidence of fifteen weeks elapsed. Four role slots distribute coordination/ethics, content/accessibility, engineering/data and QA/documentation. Names and institutional allocation are left for the project team.

6.2 The Program (Code)

The listing below is the file executed for the captured run. Shared I/O helpers are reproduced in Appendix D.

```
from evidence_common import load, result
plan = load('docs/plan.json')
assert [r['week'] for r in plan] == list(range(1, 16))
term = sum(r['term_hours'] for r in plan)
self_hours = sum(r['self_hours'] for r in plan)
assert (term, self_hours) == (30, 30)
print(f'Planned weeks={len(plan)}; term={term}h; self={self_hours}h')
print('Roles=' + str(len(set(r['role'] for r in plan))))
print('Schedule status=proposed; not an attendance record')
result(5, '15-week effort allocation closes at 60 hours')
```

6.3 Captured Run (Two-Pass Verified)

```
$ python scripts/cp05_week.py
Planned weeks=15; term=30h; self=30h
Roles=4
Schedule status=proposed; not an attendance record
CP-05 RESULT: 15-week effort allocation closes at 60 hours
```

6.4 What the Capture Proves

The plan allocates two term-work hours and two self-learning hours each week, closing at 30 + 30 hours. A school-readiness gate precedes field demonstration. If permission or content review is delayed, week 12 remains pending rather than being replaced with simulated interviews. The JSON plan can be edited and checked without silently changing the report totals.

RESULT (THE PROGRAM'S OWN LINE)

CP-05 RESULT: 15-week effort allocation closes at 60 hours

6.5 Checkpoint Questions

Q1. What happens if school engagement is delayed?

Model answer. Continue safe engineering work and report the engagement gate as unresolved. Do not represent automation as a classroom pilot.

Q2. How is effort different from money?

Model answer. The 60 hours are a proposed educational workload. Hardware, staff opportunity cost, travel and hosting still need local estimates.

6.6 Field Extensions

Assign real role owners and review dates. Record actual effort separately from this plan; track review delays and refinements with Git issues or a local issue ledger.

Chapter 7 — Feasibility: What the School Must Confirm

WEEK (TLP ROW) 7 — Feasibility analysis and solution brainstorming

EVIDENCE Solution alternatives sheet

FILES scripts/cp06_week.py; evidence/cp06_week.txt

COURSE OUTCOMES CO5 (reference alignment only)

7.1 Why This Week

A static tool was chosen to reduce runtime complexity: no server-side inference, user accounts, application database or mandatory API subscription. Existing browser hardware is an assumption to validate, not an asset inventory obtained from a school. Production dependencies differ from development dependencies: the app has none in npm, but testing uses Playwright and rebuilding data uses Python libraries.

7.2 The Program (Code)

The listing below is the file executed for the captured run. Shared I/O helpers are reproduced in Appendix D.

```
from evidence_common import load, ROOT, result
pkg = load('package.json')
prod_deps = len(pkg.get('dependencies', {}))
assert prod_deps == 0
size = sum(p.stat().st_size for p in (ROOT/'public').rglob('*')
          if p.is_file())
print(f'Production npm dependencies={prod_deps}')
print(f'Bundled site bytes={size}')
print('Required API keys=0; server database=none')
print('Hardware, staff time, internet and hosting cost=unconfirmed')
result(6, 'static feasibility measured; cost assumptions not quotations')
```

7.3 Captured Run (Two-Pass Verified)

```
$ python scripts/cp06_week.py
Production npm dependencies=0
Bundled site bytes=1310555
Required API keys=0; server database=none
Hardware, staff time, internet and hosting cost=unconfirmed
CP-06 RESULT: static feasibility measured; cost assumptions not quotations
```

7.4 What the Capture Proves

The program measures the deployed directory's actual byte count and the production dependency count. The payload includes font, digits, model vectors and map so essential learning can work after a first successful cache. Low bandwidth and browser cache eviction remain real constraints. No zero-cost field implementation or guaranteed free hosting is asserted.

RESULT (THE PROGRAM'S OWN LINE)

CP-06 RESULT: static feasibility measured; cost assumptions not quotations

7.5 Checkpoint Questions

Q1. Why does an offline-capable tool still need an initial connection?

Model answer. The HTTPS site must first download and cache its assets. Local HTTP serving is an alternative; opening index.html directly via file:// is not supported.

Q2. What costs remain if API keys are unnecessary?

Model answer. Device availability, electricity, internet, educator time, accessibility equipment, travel, maintenance and any hosting charges.

7.6 Field Extensions

Test on the actual school device and browser. Obtain resource/cost estimates, agree who updates the content, and keep a local copy for network or cache failures.

Chapter 8 — Design Week: The Engine Surface and the Page Graph

WEEK (TLP ROW) 8 — Selection of final solution and design approach

EVIDENCE Concept note / design outline

FILES scripts/cp07_week.py; evidence/cp07_week.txt

COURSE OUTCOMES CO2, CO6 (reference alignment only)

8.1 Why This Week

The app has five views: learning home, practice, device progress, district context and educator resources. The practice engine presents a short sequence, accepts a choice, offers a supportive retry or confirms success, then enables Next. It never advances on a timer. The separate digit model does not decide whether a child can progress.

8.2 The Program (Code)

The listing below is the file executed for the captured run. Shared I/O helpers are reproduced in Appendix D.

```
from evidence_common import read, result
app = read('public/app.js')
engine = read('public/engine.js')
for token in ['function home', 'function progress',
              'function district', 'function educators',
              'function practice']:
    assert token in app
for token in ['selectCards', 'summarize', 'grade', 'predictKnn']:
    assert token in engine
assert 'remember:false' in app
print('Views=5; default persistence=off; answer state=explicit')
print('Model inference separated from learner practice grading')
result(7, 'page graph and explicit engine surfaces verified')
```

8.3 Captured Run (Two-Pass Verified)

```
$ python scripts/cp07_week.py
Views=5; default persistence=off; answer state=explicit
Model inference separated from learner practice grading
CP-07 RESULT: page graph and explicit engine surfaces verified
```

8.4 What the Capture Proves

The static checks confirm named view/engine surfaces and memory-only default storage; the browser tests later verify actual behaviour. Each session contains three or five cards. A category's next session starts after its device completion count and wraps through the pack; this is deterministic rotation, not adaptive intelligence. CSV records contain date, session token, category, card, attempts and first-correct flag.

RESULT (THE PROGRAM'S OWN LINE)

CP-07 RESULT: page graph and explicit engine surfaces verified

8.5 Checkpoint Questions

Q1. Is the educator view a secure admin area?

Model answer. No. It is a public informational workspace and demonstration, with no privileged actions or private learner records.

Q2. Why separate first-try success from retries?

Model answer. A card eventually completed after help is not the same as an unaided first response. Even this distinction is only practice behaviour, not an ability or clinical measure.

8.6 Field Extensions

Observe whether instructions, focus changes, feedback and exit controls are understandable. Offer alternative tasks for visual impairments and individual sensory preferences; do not assume one interface serves every learner.

PART B

BUILD, TEST AND RESPONSIBLE HANDOVER

Chapters 9–15 · Dataset processing, district mapping, browser implementation, technical tests, pending field protocol, outcome limits and publication.

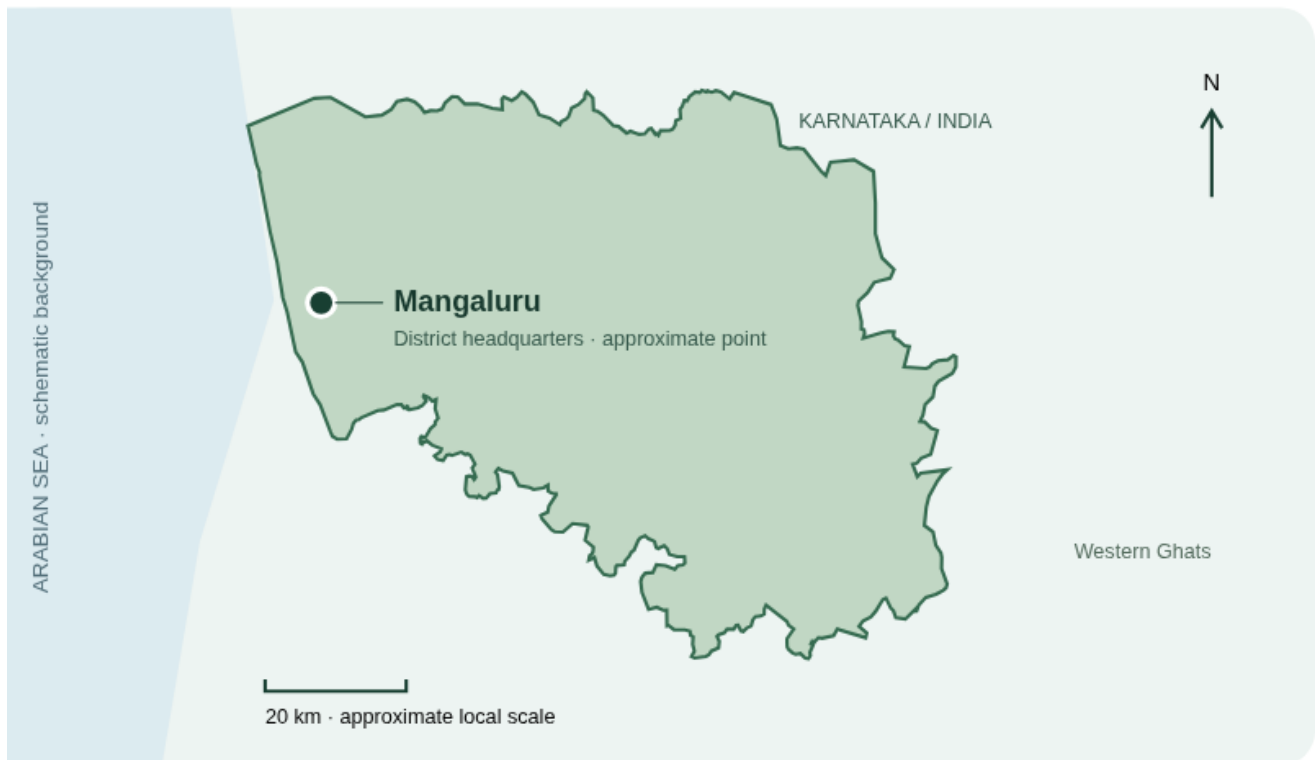


Figure B. District geometry from geoBoundaries, 2021 vintage, ODbL 1.0. Approximate Mangaluru point; not a navigation or school-service map.

Chapter 9 — The Content Pack: Secondary Digits and a District Map

WEEK (TLP ROW) 9 — Data processing / calculations / sizing / mapping

EVIDENCE Technical analysis sheet

FILES scripts/cp08_week.py; evidence/cp08_week.txt

COURSE OUTCOMES CO2, CO6 (reference alignment only)

9.1 Why This Week

The content pack contains authored everyday activities and a clearly separated optional handwriting module. The model uses UCI's original split: 3,823 training rows and 1,797 test rows [S1]. Each image is an 8×8 array of intensities from 0 to 16; Euclidean 3-nearest-neighbours voting is a transparent baseline. No test-label tuning or new learner handwriting collection occurs.

9.2 The Program (Code)

The listing below is the file executed for the captured run. Shared I/O helpers are reproduced in Appendix D.

```
from collections import Counter
from evidence_common import load, result
cards = load('public/data/lessons.json')
counts = dict(sorted(Counter(c['category'] for c in cards).items()))
m = load('public/data/model-metrics.json')
assert len(set(c['id'] for c in cards)) == 30
assert m['correct'] == sum(m['confusion_matrix'][i][i]
                           for i in range(10))
print(f'Cards by category={counts}')
print(f"3-NN correct={m['correct']}/{m['test_rows']}")
print(f"Repository test accuracy={m['accuracy']:.6f}")
result(8, 'content and confusion-matrix totals agree')
```

9.3 Captured Run (Two-Pass Verified)

```
$ python scripts/cp08_week.py
Cards by category={'colours': 5, 'digits': 10, 'life': 5, 'numbers': 5, 'shapes': 5}
3-NN correct=1758/1797
Repository test accuracy=0.978297
CP-08 RESULT: content and confusion-matrix totals agree
```

9.4 What the Capture Proves

The program validates the confusion-matrix diagonal against the saved correct count. The full accuracy comes from Python reference evaluation; browser parity is checked on 20 selected held-out examples only. The selected examples are not an independent evaluation. Ten digit cards use these secondary images; twenty everyday cards are author-written. The map is another genuine secondary-data derivative, not an invented sketch.

RESULT (THE PROGRAM'S OWN LINE)

CP-08 RESULT: content and confusion-matrix totals agree

9.5 Checkpoint Questions

Q1. Why is 97.83% not an educational outcome?

Model answer. It measures repository digit-label classification on a fixed split. It says nothing about improvement, accessibility or suitability for local learners.

Q2. What is the biggest domain limitation?

Model answer. UCI writer samples are not a validated sample of this learner population and are not Kannada numerals. The model is restricted to an educator demonstration.

9.6 Field Extensions

Review the optional low-resolution handwritten images for suitability. If collecting any new handwriting is later proposed, obtain separate approval, minimization rules and a held-out evaluation design; do not silently expand this prototype.

Chapter 10 — The Build: A Local-First Browser Tool

WEEK (TLP ROW) 10 — Prototype / model / audit / DPR / app development

EVIDENCE Working app / source repository

FILES scripts/cp09_week.py; evidence/cp09_week.txt

COURSE OUTCOMES CO6 (reference alignment only)

10.1 Why This Week

The deployed site is the public/ folder. A small Node build script hashes bundled files and writes a versioned service worker. A local Node server is provided for development; Render uses static hosting rather than that server. The app requests only same-origin bundled resources during core use and does not call model or analytics APIs.

10.2 The Program (Code)

The listing below is the file executed for the captured run. Shared I/O helpers are reproduced in Appendix D.

```
from evidence_common import read, ROOT, result
sw = read('public/sw.js')
assert 'caches.open' in sw and 'cache.addAll' in sw
assert 'node scripts/build.mjs' in read('render.yaml')
assert 'runtime: static' in read('render.yaml')
files = [p for p in (ROOT/'public').rglob('*') if p.is_file()]
print(f'Deployed public files={len(files)}')
print('Runtime=static HTML/CSS/JS; backend=none')
print('Offline prerequisite=successful initial HTTPS cache')
result(9, 'static build and Render configuration present')
```

10.3 Captured Run (Two-Pass Verified)

```
$ python scripts/cp09_week.py
Deployed public files=19
Runtime=static HTML/CSS/JS; backend=none
Offline prerequisite=successful initial HTTPS cache
CP-09 RESULT: static build and Render configuration present
```

10.4 What the Capture Proves

The build check confirms the static configuration and generated offline cache instructions. Offline use requires a complete initial secure-origin cache; browser voice services and external references are outside that guarantee. Cache entries are content-versioned. New workers activate after old tabs close, avoiding an uncontrolled mid-session replacement. A browser can still evict stored data.

RESULT (THE PROGRAM'S OWN LINE)

CP-09 RESULT: static build and Render configuration present

10.5 Checkpoint Questions

Q1. Why is there no backend start command in render.yaml?

Model answer. Render static sites publish a directory. There is no application database or API server to start.

Q2. Why can the model run without a cloud service?

Model answer. The exported neighbour vectors are small enough to bundle, and the browser computes distances locally over these repository examples.

10.6 Field Extensions

Perform a deployment smoke test on the assigned HTTPS URL. Check font loading, every activity, map, source links, CSV handling and an offline reload. Do not confuse a sandbox preview with a deployed school service.

Chapter 11 — Testing: A Truth Table for the Engine, a Browser Receipt

WEEK (TLP ROW) 11 — Testing, refinement, and implementation planning

EVIDENCE Test results / revised output

FILES scripts/cp10_week.py; evidence/cp10_week.txt

COURSE OUTCOMES CO2, CO4 (reference alignment only)

11.1 Why This Week

Testing follows three layers: deterministic content/engine checks, Chromium user-flow automation, and limited automated accessibility scans. There is also an artifact regeneration check. These layers catch engineering defects without pretending to replace classroom observation or an accessibility specialist.

11.2 The Program (Code)

The listing below is the file executed for the captured run. Shared I/O helpers are reproduced in Appendix D.

```
from evidence_common import load, read, result
browser = load('evidence/browser-results.json')
axe = load('evidence/accessibility.json')
unit = read('evidence/unit-transcript.txt')
assert browser['status'] == 'PASS' and not browser['errors']
assert '# pass 9' in unit and '# fail 0' in unit
violations = sum(len(row['violations']) for row in axe)
print('Recorded Node unit tests=9 passed, 0 failed')
print(f"Recorded Chromium checks={len(browser['checks'])} passed")
print(f'Automated axe findings across {len(axe)} views={violations}')
print('Scope=recorded automation; not full accessibility certification')
result(10, 'technical test receipts checked; no user trial inferred')
```

11.3 Captured Run (Two-Pass Verified)

```
$ python scripts/cp10_week.py
Recorded Node unit tests=9 passed, 0 failed
Recorded Chromium checks=19 passed
Automated axe findings across 4 views=0
Scope=recorded automation; not full accessibility certification
CP-10 RESULT: technical test receipts checked; no user trial inferred
```

11.4 What the Capture Proves

The receipts record nine passing Node tests and nineteen passing browser checks. Four main views have no detected axe violations in the recorded scan. Checks include retries, first-try accounting, completion, CSV, default non-retention, opt-in persistence, reset, Kannada switching, map serving, local model inference, source filtering, mobile width and offline practice. The workflow was tested in Chromium, not all browsers and assistive technologies.

RESULT (THE PROGRAM'S OWN LINE)

CP-10 RESULT: technical test receipts checked; no user trial inferred

11.5 Checkpoint Questions

Q1. Does a clean axe report mean the app is WCAG-conformant?

Model answer. No. It is a limited automated scan. Manual keyboard, screen-reader, cognitive-accessibility and individual-user evaluation remain necessary.

Q2. How did testing improve the delivery?

Model answer. The capture records include a deterministic engine suite and a corrected test-navigation wait. Map annotation placement and mobile screenshots were visually reviewed. Technical regressions can be rerun in CI.

11.6 Field Extensions

Add teacher-led task walkthroughs, actual keyboard/switch testing, screen-reader review and language verification. Record failed cases as well as successes and maintain a defect/refinement log.

Chapter 12 — The Pilot: A Protocol, Not Invented Participants

WEEK (TLP ROW) 12 — Field implementation / demonstration / awareness activity

EVIDENCE Implementation record (pending)

FILES scripts/cp11_week.py; evidence/cp11_week.txt

COURSE OUTCOMES CO4 (reference alignment only)

12.1 Why This Week

The reference book's week 12 is field implementation or demonstration. In this companion it is deliberately a protocol, because no fieldwork has occurred. The included sample practice CSV was generated by browser automation. It must never be presented as a child's session or a school pilot.

12.2 The Program (Code)

The listing below is the file executed for the captured run. Shared I/O helpers are reproduced in Appendix D.

```
from evidence_common import read, result
protocol = read('docs/FIELD_PROTOCOL.md')
for term in ['PENDING', 'consent', 'assent', 'fatigue']:
    assert term in protocol
assert 'No schools' in protocol
print('Field visits=0; interviews=0; learner trials=0')
print('Evidence delivered=blank protocol and observation form')
print('Next gate=school permission and educator content review')
result(11, 'pilot protocol exists; no fabricated participants')
```

12.3 Captured Run (Two-Pass Verified)

```
$ python scripts/cp11_week.py
Field visits=0; interviews=0; learner trials=0
Evidence delivered=blank protocol and observation form
Next gate=school permission and educator content review
CP-11 RESULT: pilot protocol exists; no fabricated participants
```

12.4 What the Capture Proves

The program verifies that the protocol is marked PENDING and contains consent, assent and fatigue considerations. Delivered evidence consists of the protocol and blank observation fields, not attendance, photographs or school signatures. Secondary resources and automated tests can support a demonstration, but cannot substitute for community involvement.

RESULT (THE PROGRAM'S OWN LINE)

CP-11 RESULT: pilot protocol exists; no fabricated participants

12.5 Checkpoint Questions

Q1. Can the team fill the implementation record from the app screenshot?

Model answer. No. A screenshot proves a technical view existed. It does not establish participation, permission, acceptance or impact.

Q2. What should stop a demonstration?

Model answer. Missing permission/assent, distress, fatigue, unsuitable content, unsupported access needs or unsafe data handling. The learner can stop without penalty.

12.6 Field Extensions

Obtain approval and agree on an untimed, educator-supported activity. Record only necessary non-identifying observations. Keep signed permissions outside the public Git repository and do not upload identifiable photos.

Chapter 13 — Impact and the DPR: Numbers That Can Be Defended

WEEK (TLP ROW) 13 — Documentation of outcomes, feedback, and impact

EVIDENCE Draft report / impact limits

FILES scripts/cp12_week.py; evidence/cp12_week.txt

COURSE OUTCOMES CO3, CO4, CO6 (reference alignment only)

13.1 Why This Week

The detailed project report must distinguish an output from an outcome. A working app, content pack, sourced map, model, tests and deployment files are engineering outputs. Improved engagement, independence or learning would be social/educational outcomes requiring an approved evaluation. None has been measured here.

13.2 The Program (Code)

The listing below is the file executed for the captured run. Shared I/O helpers are reproduced in Appendix D.

```
from evidence_common import load, result
m = load('public/data/model-metrics.json')
counts = len(load('public/data/lessons.json'))
b = load('evidence/browser-results.json')
print(f'Technical output={counts} bilingual cards')
print(f'Automation checks={len(b['checks'])}')
print(f'Repository errors={m['test_rows']-m['correct']}')
print('Measured learner improvement=NOT MEASURED')
print('Societal benefit=plausible aim, not validated impact')
result(12, 'engineering outputs separated from social-outcome claims')
```

13.3 Captured Run (Two-Pass Verified)

```
$ python scripts/cp12_week.py
Technical output=30 bilingual cards
Automation checks=19
Repository errors=39
Measured learner improvement=NOT MEASURED
Societal benefit=plausible aim, not validated impact
CP-12 RESULT: engineering outputs separated from social-outcome claims
```

13.4 What the Capture Proves

The ledger recomputes pack size, automation count and repository model errors from saved artifacts. It explicitly leaves learner improvement unmeasured. Future comparisons must state denominators, assistance, missing sessions, task changes and sampling limitations. Repeated practice without a control or comparable baseline cannot establish a causal effect.

RESULT (THE PROGRAM'S OWN LINE)

CP-12 RESULT: engineering outputs separated from social-outcome claims

13.5 Checkpoint Questions

Q1. How is SDG 4 supported without an impact percentage?

Model answer. The design aims at inclusive educational access. This is an alignment argument, not evidence of a changed SDG indicator.

Q2. What should a final report say if no pilot happens?

Model answer. Report the technical results fully, explain the unresolved fieldwork and avoid claiming school acceptance, learning gains or full rubric completion.

13.6 Field Extensions

If approved, agree on meaningful individual goals and descriptive measures with an educator before evaluation. Report limitations and negative observations alongside any benefit.

Chapter 14 — The Evidence Check: Recompute Everything, Compare Bytes

WEEK (TLP ROW) 14 — Presentation preparation and viva practice

EVIDENCE Presentation outline / reproducibility receipts

FILES scripts/cp13_week.py; evidence/cp13_week.txt

COURSE OUTCOMES CO2, CO4 (reference alignment only)

14.1 Why This Week

Reproducibility is a discipline, not a decorative hash. The evidence manifest contains SHA256 fingerprints for bundled source/site artifacts. The verification script checks hashes, regenerates derived data and the service worker, then checks again. The collector separately runs each week program twice and rejects nonzero exits, stderr or unequal stdout.

14.2 The Program (Code)

The listing below is the file executed for the captured run. Shared I/O helpers are reproduced in Appendix D.

```
from evidence_common import load, sha, result
manifest = load('evidence/artifact-manifest.json')
for filename, expected in manifest.items():
    assert sha(filename) == expected, filename
print(f'Verified source/site artifact hashes={len(manifest)}')
print('Lesson pack SHA256=' + sha('public/data/lessons.json'))
result(13, 'committed artifacts match their recorded fingerprints')
```

14.3 Captured Run (Two-Pass Verified)

```
$ python scripts/cp13_week.py
Verified source/site artifact hashes=22
Lesson pack SHA256=771e6fb7fbc39731930c5a11d1dc271fb6567af927b8059b27b1da6126517495
CP-13 RESULT: committed artifacts match their recorded fingerprints
```

14.4 What the Capture Proves

The two-pass claim applies to the fifteen deterministic week transcripts. Browser screenshots, timing fields and test durations are not claimed byte-identical. The report quotes the actual program listings and captured transcripts. A source hash proves equality to the recorded bytes; provenance still depends on the source register, licence and original repository.

RESULT (THE PROGRAM'S OWN LINE)

CP-13 RESULT: committed artifacts match their recorded fingerprints

14.5 Checkpoint Questions

Q1. What happens after legitimate content changes?

Model answer. Review and test the change, rebuild, intentionally refresh the artifact manifest, recapture evidence and regenerate the report. Commit the reviewable difference.

Q2. Which presentation structure is easiest to defend?

Model answer. Problem and limits; local context; sources; architecture; working demonstration; technical results; pending field validation; deployment and handover.

14.6 Field Extensions

Rehearse the demo from a clean browser and a cached offline copy. Be ready to open raw data, source metadata, code, tests and the field-status note rather than recite unsupported statistics.

Chapter 15 — The Viva Ledger: Handover with Git and Render

WEEK (TLP ROW) 15 — Final presentation, submission, and reflection

EVIDENCE Final report, deployment handover, viva

FILES scripts/cp14_week.py; evidence/cp14_week.txt

COURSE OUTCOMES CO1, CO2, CO3, CO4, CO5, CO6 (reference alignment only)

15.1 Why This Week

Handover consists of a Git repository, documentation, artefacts and a realistic publication path. Local Git history has been created. The supplied Render Blueprint points to the static directory and needs no secret. Publishing a remote repository and creating the Render service require the owner's accounts; those actions have not been performed.

15.2 The Program (Code)

The listing below is the file executed for the captured run. Shared I/O helpers are reproduced in Appendix D.

```
from evidence_common import read, load, result
for filename in ['render.yaml', 'docs/DEPLOYMENT.md',
                 '.github/workflows/ci.yml', 'LICENSE']:
    assert read(filename).strip()
m = load('public/data/model-metrics.json')
print('Viva: model metric evaluates digits, never learner ability')
print(f"Viva: original source test rows={m['test_rows']}")
print('Handover=Git source + static Render Blueprint + guide')
print('Remote Git push and Render account deployment=NOT PERFORMED')
result(14, 'handover files verified; remote actions explicitly pending')
```

15.3 Captured Run (Two-Pass Verified)

```
$ python scripts/cp14_week.py
Viva: model metric evaluates digits, never learner ability
Viva: original source test rows=1797
Handover=Git source + static Render Blueprint + guide
Remote Git push and Render account deployment=NOT PERFORMED
CP-14 RESULT: handover files verified; remote actions explicitly pending
```

15.4 What the Capture Proves

The program verifies required handover files and prints the pending remote status. The deployment guide separates local serving, Git publication, Render configuration, smoke testing and rollback. The Git bundle preserves delivered history without relying on a hosted remote. A new origin changes browser storage; progress is not a centrally backed-up record.

RESULT (THE PROGRAM'S OWN LINE)

CP-14 RESULT: handover files verified; remote actions explicitly pending

15.5 Checkpoint Questions

Q1. What is the final honest claim?

Model answer. A working, technically tested, sourced and deployable bilingual prototype with a continuation-style report. School approval, educator validation and public deployment are still pending.

Q2. Who owns ongoing safety and maintenance?

Model answer. The adopting institution must name a maintainer and educator reviewer, control content changes, protect any local records and periodically recheck device and source assumptions.

15.6 Field Extensions

Publish using your own Git/Render account, test the final URL, obtain school review and record an actual handover date. Use Git tags for reviewed releases and keep credentials out of source control.

APPENDICES

Appendix A — Quick Reference, Fingerprints & Debug Log

A.1 Pack, model and map

public/data/lessons.json

```
771e6fb7fbc39731930c5a11d1dc271fb6567af927b8059b27b1da6126517495
```

public/data/dakshina-kannada.geojson

```
7ff77212aa67e679169958d251cdda0a9a95e8e5f9a394d49d15185c60dfca3f
```

public/models/knn.json

```
2c88d509d0c8f8f595a78911baf89084ceabdbfd384e17c6f2072367b3c6b05e
```

All 22 source/site artifact hashes, including font and raw UCI files, are in evidence/artifact-manifest.json. The manifest excludes logs, screenshots and the report to avoid self-referential hashes. Refresh it intentionally only after review.

A.2 Actual capture and verification receipts

```
RESULT: 15 week programs passed; two captures each; byte-identical stdout
```

```
UCI original split: train=3823, test=1797
3-NN: 1758/1797 correct; accuracy=0.978297
District: Dakshina Kannada; boundary year=2021; geometry=Polygon
Bounds: 74.7603, 12.4612, 75.6850, 13.1844
Content: 30 bilingual cards / 5 categories; Kannada review pending
Build OK: 18 bundled files; offline cache sahaya-fb0bd8e4636c3daf
RESULT: 22 artifact hashes verified before and after regeneration
```

A.3 Debug and review notes

During construction, the geoBoundaries download initially returned a Git-LFS pointer rather than geometry; the pinned media URL was used. An over-broad district-name filter matched unrelated “Dakshin” districts; the final filter requires exact “Dakshina Kannada”. A browser test initially read progress before hash navigation rendered; an explicit active-route wait corrected the test. These are development corrections, not field incidents.

Map annotation placement was adjusted to avoid clipping. Kannada uses a bundled font. Learner controls and source filtering were exercised in Chromium; native Kannada language quality and individual assistive access remain unverified.

A.4 Dataset and licence register

[S1] UCI digits — E. Alpaydin and C. Kaynak, 1998

CC BY 4.0; original split, 8×8 counts. DOI: 10.24432/C50P49
<https://archive.ics.uci.edu/dataset/80/optical+recognition+of+handwritten+digits>

[S2] geoBoundaries India ADM2 — Pathways Data Pvt. Ltd., lgdirectory.gov.in

ODbL 1.0; 2021 vintage; commit 9469f09; one extracted feature
<https://www.geoboundaries.org/api/current/gbOpen/IND/ADM2/>

[S3] Dakshina Kannada District Administration

Official factual district context; no endorsement
<https://dk.nic.in/en/about-district/>

[S4] Twemoji v14.0.2 — Twitter and contributors

CC BY 4.0 graphics; 4 unmodified bundled SVGs
<https://github.com/twitter/twemoji/tree/v14.0.2>

[S5] scikit-learn 1.6.1 — KNeighborsClassifier

BSD-3-Clause; executed 3-NN reference implementation
<https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html>

[S6] AI4Bharat IndicTrans2 En-Indic distilled 200M

Optional only; model card lists MIT and Kannada; not downloaded/executed
<https://huggingface.co/ai4bharat/indictrans2-en-indic-dist-200M>

[S7] Noto Sans Kannada — Google Fonts / Noto contributors

SIL OFL 1.1; licence and font bundled
<https://github.com/google/fonts/tree/main/ofl/notosanskannada>

[S8] Render Blueprint specification

Static service configuration reference; account/terms apply
<https://render.com/docs/blueprint-spec>

Reviewed/retrieved 20 September 2026. Modification notices, full pinned download URL, attribution and finder links are in docs/SOURCES.md. The map derivative GeoJSON retains ODbL; the project MIT licence does not override data/asset licences.

A.5 Data dictionary and model evaluation

Artifact	Fields / semantics
optdigits.tra / .tes	64 integers 0..16; column 65 = digit class 0..9
lessons.json	id, category, bilingual prompt, visual kind/value, choices, answer, source
district GeoJSON	FeatureCollection; one feature; shapeName, shapeID, geometry coordinates
model JSON	k=3; x=training vectors; y=training labels
Practice CSV	date, session, category, card, attempts, firstCorrect; no names or diagnoses

Confusion matrix: rows are true digit classes; columns are predicted classes. The following values are read from the saved full-split reference evaluation, not typed from a model description.

True / Pred	0	1	2	3	4	5	6	7	8	9
0	178	0	0	0	0	0	0	0	0	0
1	0	180	0	0	0	0	1	0	1	0
2	0	4	173	0	0	0	0	0	0	0
3	0	0	0	181	0	0	0	1	1	0
4	0	2	0	0	178	0	0	0	1	0
5	0	0	0	1	1	179	0	0	0	1
6	0	0	0	0	0	0	181	0	0	0
7	0	0	0	0	0	0	0	172	1	6
8	0	9	0	1	0	0	0	0	162	2
9	0	0	0	4	0	1	0	0	1	174

Accuracy = diagonal sum / total = 1758 / 1797 = 0.978297162. Errors = 39. No learner outcome is measured. Browser parity is checked on 20 demonstrative examples only; neighbour ties may differ across implementations.

Appendix B — Report Format, Rubrics & Submission

B.1 The report the rubric reads

Problem relevance: Chapters 1–3 establish the prescribed topic, context and the honest secondary/primary distinction.

Engineering and feasibility: Chapters 4–8 define requirements, design choices, workload and unconfirmed resource assumptions.

Stakeholder interaction: Chapters 3 and 12 supply a protocol but no executed school engagement. This rubric dimension cannot be treated as fully evidenced.

Documentation and problem-solving: Source register, real code, tests, artefacts and reproducible captures support the engineering claims.

Communication and viva: Chapters 14–15 and Appendix C provide a defensible demonstration outline.

Societal contribution: Intended inclusive-education benefit is explained; actual impact and sustainability arrangements remain to be validated.

B.2 Submission checklist

Item	Status
PDF / editable DOCX companion	Included
Source, raw data, map and model artifacts	Included with licences and hashes
Fifteen two-pass week captures	Included
Node/Chromium/axe receipts	Included; automation only
Git history / Render Blueprint / CI	Included; accounts not connected
School permission / requirement sign-off	Pending
Kannada educator review / access validation	Pending
Field attendance / feedback / impact evidence	Not collected; do not fabricate
Team details and guide approval	To be supplied

The requested table row appears in the front matter and docs/REQUIREMENTS.md. Do not assign yourself full rubric marks based only on a working prototype. Assessment remains with the authorized committee.

B.3 Implementation record — blank for authorized use

Field	Entry
School / permission reference (keep private)	Pending
Educator reviewer / date	Pending
Device / browser / access adaptations	Pending
Chosen task / language / agreed goal	Pending
Consent and assent checked	Pending
Observation / assistance / fatigue	Pending
Defects / refinements / review decision	Pending
Handover / maintainer / next review	Pending

Keep signed forms and any identifying records out of the public source repository. The sample CSV in evidence/ was produced by automation and is not a participant record. Clear shared-browser progress according to the school's agreed practice.

B.4 Outcome reporting template

Report who was invited and who participated only after approval; disclose missing observations and assistance. Use comparable tasks and denominators. Explain why any apparent change could reflect practice, prompting, selection or task differences. Without approved fieldwork, write “not measured” for learner improvement and “pending” for acceptance.

Appendix C — Viva Voce Bank

Q1. What was actually built?

Ans. A static bilingual learning tool with 30 cards, local practice records, sourced district context and an optional educator digit-model demonstration.

Q2. Which dataset really enters the app?

Ans. UCI digit examples become ten handwriting cards and the model demo. The geoBoundaries feature becomes the district map. Authored everyday cards are not mislabelled as secondary survey data.

Q3. How is 97.83% computed?

Ans. 1,758 correct classifications divided by 1,797 original UCI test rows. It is a repository classification metric, never a learning gain.

Q4. What makes it an open-source model implementation?

Ans. The reference uses scikit-learn KNeighborsClassifier under BSD-3-Clause; the original local browser equivalent is supplied as MIT code. UCI vectors retain CC BY attribution.

Q5. Why not call IndicTrans2 during each lesson?

Ans. It is not needed for fixed reviewed content. A static reviewed pack avoids remote inference, runtime variability and extra hosting; IndicTrans2 is an optional future candidate only.

Q6. Does progress belong to an identified learner?

Ans. No. It is anonymous device-level practice, memory-only by default. Opt-in storage can mix multiple users and is not a secure student database.

Q7. How do I publish it?

Ans. Push the supplied repository through your account, connect it to Render, apply render.yaml, and verify the HTTPS site. Those remote account actions remain pending.

Q8. What is still missing for community acceptance?

Ans. Real school requirements, language/access review, permission, learner assent/consent procedures, a supervised trial and feedback. Secondary data cannot replace these.

C.2 Presentation sequence

1. Prescribed topic and unmet validation gate. 2. Dakshina Kannada map and school-context questions. 3. Secondary-source provenance and licences. 4. Bilingual app walkthrough. 5. Local progress and privacy controls. 6. Model metric and limits. 7. Technical tests and reproducibility. 8. School review, Git/Render handover and next actions. This is an outline, not a fabricated record of a delivered presentation.

Appendix D — Reproducing This Edition, Git & Render

D.1 Reproduction procedure

```
# From the repository root; Python 3.13 / Node 20+ recommended
pip install -r requirements.txt
python scripts/prepare_data.py
python scripts/make_lessons.py
node scripts/build.mjs
python scripts/verify_artifacts.py
npm ci
npm test
npx playwright install --with-deps chromium
# Start in another terminal: node server.mjs
npm run test:browser
python scripts/collect.py
# PDF/DOCX generation also needs requirements-report.txt
pip install -r requirements-report.txt
python scripts/make_report.py
```

Initial package installation needs internet; data regeneration does not after dependencies and the bundled raw data are present. Browser checks need a running local server at port 3000. DOCX/PDF generation uses local DejaVu fonts on Linux; install fonts-dejavu-core or adjust font paths on another operating system.

D.2 Shared evidence helper — exact listing

```
"""Shared deterministic I/O for the fifteen week-led checks."""
import hashlib, json, pathlib
ROOT=pathlib.Path(__file__).resolve().parents[1]
def read(path): return (ROOT/path).read_text()
def load(path): return json.loads(read(path))
def sha(path): return hashlib.sha256((ROOT/path).read_bytes()).hexdigest()
def result(week, text): print(f'CP-{week:02d} RESULT: {text}')
```

D.3 Artifact updates

After approved edits, rebuild and rerun tests. Regenerate evidence/artifact-manifest.json with sorted paths and SHA256 bytes for every file under public/ and data/raw/. Review the changes, rerun verify_artifacts.py and collect.py, regenerate this report, and commit all related artifacts together. Do not refresh hashes just to hide a failed integrity check.

D.4 The collector — exact listing

```

"""Run all week checks twice; persist only byte-identical, stderr-free receipts."""
import os, pathlib, subprocess, sys
R=pathlib.Path(__file__).resolve().parents[1]
files=sorted((R/'scripts').glob('cp??_week.py'))
assert len(files)==15
for f in files:
    runs=[]
    for _ in range(2):
        p=subprocess.run([sys.executable, str(f)], cwd=R, capture_output=True,
                          env={**os.environ, 'PYTHONHASHSEED': '0'})
        assert p.returncode==0 and not p.stderr, (f.name, p.stderr.decode())
        runs.append(p.stdout)
    assert runs[0]==runs[1], f.name+' is nondeterministic'
    target=R/'evidence'/f'{f.stem}.txt'
    target.write_bytes(f'$ python scripts/{f.name}\n'.encode()+runs[0])
print('RESULT: 15 week programs passed; two captures each; byte-identical stdout')
(R/'evidence/capture-summary.txt').write_text('RESULT: 15 week programs passed; two
captures each; byte-identical stdout\n')

```

D.5 Render Blueprint — exact listing

```

services:
- type: web
  name: sahaya-learning
  runtime: static
  buildCommand: node scripts/build.mjs
  staticPublishPath: ./public
  autoDeployTrigger: commit
  headers:
  - path: /*
    name: X-Content-Type-Options
    value: nosniff
  - path: /*
    name: Referrer-Policy
    value: no-referrer
  - path: /sw.js
    name: Cache-Control
    value: no-cache

```

Render publishes public/ as a static site. Python packages and the local Node server are not runtime hosting dependencies. No environment secret is required. Check current provider quotas, pricing and terms before creating a service.

D.6 Git publication and Render handover

```
# The delivered project already has local commits.
git status
git log --oneline
# Replace the placeholder with your own empty remote repository:
git remote add origin https://github.com/YOUR_ACCOUNT/sahaya-learning.git
git push -u origin main

# To restore preserved history from the standalone bundle:
git clone sahaya-history.bundle sahaya-learning
```

In Render choose New → Blueprint, connect your repository and review render.yaml. Alternatively create a Static Site with build command node scripts/build.mjs and publish directory public. The hash-based routes do not need a rewrite. Do not configure a database, gunicorn or an application start command.

D.7 Acceptance and rollback

Open the assigned HTTPS URL and complete a session in both languages. Check reset and CSV export; inspect local model and district map; allow the service worker to cache, reopen, and test offline. Review Kannada and actual device access with an educator before classroom use. For updates, close old tabs and reopen to allow a waiting worker to activate.

Use Git tags and Render deployment history for rollback. Reverting a faulty commit and rebuilding changes the content-derived cache key. Browser progress is origin-local, can be cleared and is not a cloud backup. No Git remote or Render service was created by this delivery.

D.8 Final reflection

The engineering aim has been converted into an inspectable prototype, not a claim of community impact. The next meaningful step is not a larger model: it is an authorized conversation with the special school, careful language/access review, and a small supported trial whose outcomes are honestly recorded.